SUMAN MADIPEDDI

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SUMMARY

AI/ML Engineer with **2** years of experience building **production systems** from concept to launch. Specialized in **Multi agents, GenAI,** computer vision and **full stack**. Seeking a founding role to leverage deep technical expertise and drive strategic growth.

TECHNICAL SKILLS

Languages & Frameworks: Python, Swift, C++, TypeScript, JavaScript, Node, Django, FastAPI, React, Next.js, Tailwind CSS, GraphQL

Deep Learning: PyTorch, TensorFlow, Keras, JAX, CoreML, TensorRT, ONNX, OpenCV, Stable Diffusion, CUDA, CLIP, YOLO

AI/ML: LangChain, LangGraph, A2A, mcp-use, Fine-tuning, PEFT, QLoRA, RLHF, vLLM, Pinecone, Qdrant, RAG, Conversational Agents, XGBoost, HuggingFace, VLMs, VLAs, OpenAI API, Anthropic, Gemini, Streamlit, Scikit-Learn, A/B Testing, LiveKit, Twilio

Data & MLOps: Pandas, NumPy, ETL, SQL, NoSQL, MongoDB, Supabase, PostgreSQL, Neo4j, Redis, SageMaker, Bedrock

Cloud & DevOps: AWS (EC2, Lambda, S3, EKS, ECS, DynamoDB), GCP Vertex AI, Azure, Docker, Kubernetes, GitHub Actions

WORK EXPERIENCE

AI/ML Software Engineer: Stealth AI startup, San Jose **Oct 2025 - Present**

- Architected **multi agent** orchestration document intelligence converts unstructured PDF into **knowledge graph** with LangGraph.
- **Engineered a pipeline** using Unstructured and multimodal LLMs to extract text, tables, images and generate **schema** aware Cypher queries, **evaluating agents**, producing nodes with **visual embeddings,** diagnostics, and conditional remediation logic.
- Achieved **96% extraction accuracy** and built an **intent** based **retrieval copilot** for troubleshooting workflows to diagnose.

Founding AI Engineer: Minor Chores, USA **Aug 2024 - Oct 2025**

- Designed and **Fine-tuned** LLaMA-3.1-8B with **PEFT LoRA** (rank-16) and INT8/FP8, 4-bit quantization on 4K+ proprietary examples, reducing inference costs **by 78%** while maintaining **89% accuracy**, deployed with **vLLM** achieving 120ms P95 latency.
- Developed RAG chatbot with Vertex AI, **Pinecone** Vector DB serving over 3000 queries per month. Built **ML recommendation model** retrieving real-time users by **SQL queries**, and using location, task, improving task assignment accuracy by 86%.
- Led the development of **React** app (iOS and Android) deployed on AWS ECS Fargate, **auto scalable zero downtime** and real-time messaging, payment integration, and push notifications, **shipped to 1000+ users** in 2 releases.
- **Analyzed** customer retention data, boosting engagement **by 30%** and reducing latency by **40%** in the first product launch.

PROJECT EXPERIENCE

Multi-Agent Mobile QA Automation System (Agent S3, bBoN rollouts, VLM) **Dec 2025**

- Designed a **multi-agent mobile QA system** to automate end-to-end testing of Android applications, addressing limitations of rule-based UI testing by enabling LLM-driven reasoning over **GUI state**, screenshots, and **test intent**.
- Implemented Supervisor**, Planner, Executor** architecture using **Simular Agent S3**, integrating **ADB device control**, with screenshot and **vision-grounded VLM** (UI-TARS-1.5-7B) to plan and execute UI actions across **10+** test steps per scenario.
- **Validated** the system on multiple Obsidian app workflows, executing multi-step actions with **90%** execution accuracy, and identifying the **exact failure step and root cause** through UI state and vision-based verification

Vision Agents Real Time Multi-Camera Surveillance **Dec 2025**

- Built a vision agent platform processing **multi camera RTSP streams at 30 FPS** using **GStreamer, OpenCV**, enabling real-time **YOLOv11** detection, tracking, and alerting.
- **Optimized GPU** inference by ONNX on **TensorRT** with RTX 3050 for **low latency, high throughput**. Integrated a Vision Language Model to **predict** and track user behavior from natural language queries, enabling text driven visual reasoning and actions.
- **Containerized and deployed fully on-prem**, delivering a scalable, secure **vision agents system** with zero data egress.

On-Device Real-Time Gesture Recognition System **June 2025**

- Created a real-time, **on-device** gesture and voice recognition system dynamically overlays emojis and confetti effects on live video streams, enhancing user interaction for video calls and streaming platforms.
- Trained gesture classification models with **Flax/JAX**, optimized inference pipeline with **ONNX Runtime** and **CUDA,** achieving low-latency inference and robust performance across **17-**hand gestures. Synthesized **Whisper**-based voice command recognition for spontaneous, conversational emoji and effect triggers, supporting multimodal interaction.
- Architected a modular**, privacy-first overlay pipeline** using OpenCV and Pillow enabling integration with **OBS and RTSP.**

EDUCATION

M.S. Robotics and Autonomous Systems **May 2025**

Arizona State University, Tempe, Arizona 3.8 GPA

- **Relevant Coursework:** Artificial Intelligence, Applied AI and ML, Reinforcement Learning and Perception in Robotics